Banking Loan Risk Analysis Application

System Design Document & Architecture Specification

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1. Problem Statement & Business Objective

Banks and financial institutions process tens of thousands of credit lines and retail loan accounts. Credit risk analysts face significant operational friction from manual, spreadsheet-based monitoring. The objective of this project is to build an automated, robust full-stack loan risk analysis system featuring an end-to-end ETL pipeline, high-performance relational storage in MySQL, explainable credit risk bands (Low, Medium, High), automated manual review queue triggers, role-based REST APIs, and an interactive Streamlit portal.

2. High-Level Architecture Design

The system adopts a modular microservice-style design partitioned into four decoupled layers:

- **Data Ingestion & ETL Layer:** Ingests UCI Credit Card Clients dataset (30,000 records). Runs 8 automated data-quality audits (ranges, domain codes, anomalies) and performs cryptographic MD5 record-level reconciliation.
- **Relational Data Mart (MySQL):** 3NF normalized schema for customers, credit_profiles, monthly bills, and payment records, augmented with pre-aggregated materialized KPI tables for instant dashboard performance.
- **Risk Scoring & Underwriting Rules Engine:** Computes composite risk scores (0-100), maps clients to Low, Medium, High bands, and flags anomalous accounts for mandatory manual underwriter review.
- **REST API & Presentation Layer:** FastAPI ASGI backend exposing paginated, sorted, and role-protected endpoints, consumed by a multi-tab Streamlit dashboard.

3. Technology Selection & Trade-Offs

- **FastAPI vs Flask:** FastAPI was selected for asynchronous ASGI performance, native OpenAPI documentation, and strict type validation using Pydantic V2.
- **MySQL with SQLite Auto-Fallback:** Designed for production MySQL with connection pooling, indexes, and foreign keys, with graceful local fallback ensuring seamless developer setup.
- **Pandas & SQLAlchemy ORM:** Efficient vector-based cleaning combined with type-safe ORM entities and transactional commits.
- **Streamlit:** Rapid interactive web UI enabling credit analysts and administrators to evaluate what-if scenarios and manage review queues without frontend boilerplate.

4. Business Rules & Risk Band Thresholds

- **High Risk Band:** Max payment delay ≥ 2 months, OR credit utilization ≥ 85% with repayment ratio < 20%, OR documented historical default next month, OR composite score ≥ 65.
- **Medium Risk Band:** Max payment delay = 1 month, OR utilization between 60% and 85%, OR repayment ratio between 20% and 50%, OR score 35-64.
- **Low Risk Band:** Max payment delay ≤ 0, utilization < 60%, repayment ratio ≥ 50%, and composite score < 35.
- **Manual Underwriter Review Triggers:** High credit exposure (> NT00,000) with delay; overleveraged accounts (> 90% utilization, < 15% repayment); chronic delinquency across 3+ cycles; large balance (> NT0,000) with zero recent repayment.

5. Challenges and Design Decisions

• *Challenge:* Unnormalized monthly columns in source dataset (BILL_AMT1..6, PAY_AMT1..6, PAY_0..6). *Decision:* Transformed into normalized relational tables (bills and payments) with foreign keys and compound indexes for fast time-series trend queries.

• *Challenge:* Performance overhead on analytical queries over 30,000 records. *Decision:* Implemented pre-aggregated summary tables (segment_kpi_summaries) dynamically refreshed during ETL to provide sub-millisecond KPI response times.